

ThaiEditBench: A Mechanism-Typed, Length-Stratified Benchmark for Thai Orthographic Error Correction

Nutchanon Yongsatianchot Piyalitt Ittichaiwong Kanyakorn Veerakanjana

Faculty of Medicine Siriraj Hospital, Mahidol University, Bangkok, Thailand

{nutchanon.yon, piyalitt.itt, kanyakorn.vee}@mahidol.ac.th

Abstract

Thai is written without spaces between words, and its spelling depends on tone marks, silent consonants, and the transliteration of many loanwords. Each of these is a common source of misspellings. We present **ThaiEditBench**, a benchmark for correcting Thai spelling errors at the character level. It has three components: (i) a **six-class taxonomy** that types each error by the part of the spelling that goes wrong, plus an etymology tag; (ii) **three length tiers** (sentence, paragraph, and page) that test whether a model stays accurate over long, mostly correct text; and (iii) a **contamination-robust construction**, in which naturally attested misspellings are *injected* into clean text, so the corrupted passages do not appear verbatim in the source corpora. Because the errors are injected, the gold corrections are exact, and a blind two-annotator study measures how reliably the taxonomy can be applied. Grading is at the level of Thai character clusters, with span-based matching adapted from ERRANT. We evaluate **16 models** and find that (1) under the benchmark’s conditions, a model that locates an error almost always fixes it correctly, so what separates models is *detection*; (2) the evaluated Thai-adapted models do not lead the strongest general models; and (3) in long, mostly correct passages, weaker models accumulate over-edits that pull their scores far below their sentence-level results. On real errors mined from Wikipedia edit history, scores are somewhat lower, but the model ranking largely holds.

1 Introduction

Thai spelling goes wrong in ways that are specific to its script. Writers choose the wrong tone mark, add or drop the mark that silences a consonant, confuse consonants or vowel forms that sound alike, reorder the letters of a consonant cluster, and transliterate loanwords inconsistently. For instance, สังเกต (‘to observe’) is often written สังเกตู, with an extra vowel mark. These are **character-level** errors, and

because Thai has no spaces between words, they are hard to locate and score with word-based tools. However, the resources to study them are missing. Grammatical error correction (GEC) benchmarks are largely English (Bryant et al., 2017, 2019; Napoles et al., 2017), and the closest character-level task is Chinese spelling check (CSC) (Tseng et al., 2015; Liu et al., 2011). Thai error-annotated resources cover social-media, informal, or learner text (Lertpiya et al., 2020; Nakwijit and Purver, 2022). Document-level GEC studies correction over long context (Yuan and Bryant, 2021), but no benchmark tests how Thai spelling correction holds up over long passages that contain few errors.

This work addresses this gap with a formal-register benchmark that types each error by how it arises and that limits contamination. We score at the level of the Thai Character Cluster (TCC) (Theeramunkong et al., 2000), a group of characters, such as a consonant with its vowel and tone marks, that Thai writing rules do not allow to be split. Our work contributes the following. (1) **ThaiEditBench**, a character-level benchmark for Thai spelling correction. It types each error with one of six *mechanism* classes, the part of the spelling that goes wrong, plus an etymology tag, and it has **three length tiers** (sentence, paragraph, and page). It is **contamination-robust by construction**: we inject naturally attested misspellings into clean text, so the corrupted passages are new and the gold corrections are exact. A blind annotation study measures how reliably the taxonomy can be applied (Krippendorff’s $\alpha = 0.75$). (2) An **ERRANT-style edit-typer for Thai**. It splits the difference between an input and a correction into TCC-level edits and assigns each edit one of the six classes where one applies. This lets us report **span-F0.5** separately for each error type, as ChERRANT does for Chinese (Zhang et al., 2022). We use it to score and diagnose systems on this benchmark, not as a general Thai grammatical error classifier. (3) A

16-model evaluation. Models that find an error almost always fix it, so detection separates them. The evaluated Thai-adapted systems do not lead, weaker models accumulate over-edits in long text, and the ranking largely holds on naturally occurring errors (Spearman $\rho = 0.88$). We release the code, data, grader, edit-typer, and variant map at <https://github.com/yongsa-nut/ThaiEditBench>.

2 Related Work

GEC and CSC. English grammatical error correction is evaluated with span-based metrics. These metrics represent a correction as a set of edits, each a span of the input and its replacement, and credit a system for each gold edit it reproduces with the same replacement. The M^2 scorer matches a system’s edits against gold edits (Dahlmeier and Ng, 2012). ERRANT (the ERRor ANnotation Toolkit) extracts the edits automatically by aligning the input with each correction, assigns each edit an error type, and reports span-F0.5, an F-score over matched edits that weights precision twice as much as recall (Bryant et al., 2017). GLEU instead measures fluency (Napoles et al., 2015). We adapt ERRANT-style matching to TCC units (Section 4.2). Character-level Chinese spelling check is the closest task, scoring position-based detection and correction F1 without word boundaries (Tseng et al., 2015; Wu et al., 2013; Yu et al., 2014), with confusion-set construction (Liu et al., 2011) and the MuCGEC/ChERRANT dual-granularity scorer (Zhang et al., 2022). ERRANT has been ported to Arabic (Belkebir and Habash, 2021) and Korean (Yoon et al., 2023), but not Thai.

Thai NLP. We build on PyThaiNLP for tokenization and dictionary tooling (Phatthiyaphaibun et al., 2023) and on TCC segmentation (Theeramunkong et al., 2000). Prior Thai spelling correction targets social text (Lertpiya et al., 2020, 2023), including the VISTEC-TP-TH-2021 Twitter corpus with misspelling and word-segmentation annotation (Limkonchotiawat et al., 2021), and Nakwijit and Purver (2022) study misspelling semantics.

3 The ThaiEditBench Benchmark

3.1 Task

Given a Thai passage containing character-level orthographic errors, return the passage with those errors corrected and **nothing else changed**: no rewriting, re-spacing, numeral, or punctuation edits. We score under a strict **minimal-edit contract**, where

any change to a correct span is an *over-edit* and is penalized.

3.2 Mechanism taxonomy

We define six orthographic mechanism classes, each carrying an **etymology tag**: (1) tone mark, (2) consonant-pair, (3) silent-consonant, (4) vowel, (5) cluster or metathesis, and (6) loanword transliteration. Worked examples per class are in Appendix A. We check canonical forms against the Royal Institute Dictionary (Royal Institute, 2013) (Appendix C). Class 6 is governed by a **loanword-absorption rule**: only *contemporary* foreign loans are class 6, while long-integrated borrowings (Tamil, Khmer, Persian, and Sanskrit–Pali doublets) count as native.

3.3 Construction

- **(a) Confusion-set injection.** The confusion set holds 1,044 wrong→correct pairs across the six classes. Every wrong form is a **naturally attested misspelling**, harvested from the Thai Wiktionary appendix of commonly misspelled words and from fixes made by a Thai Wikipedia spelling-cleanup bot; a rule-based classifier assigns each pair a mechanism class (96.5% agreement with 90 hand-labeled pairs; Appendix C). Only the placement is synthetic: we inject pairs into *new clean text* by matching **whole tokens**, which protects short valid forms, and record each error’s TCC span, class, and etymology. Applying the gold edits restores the clean original for 100% of items, so the gold is exact.
- **(b) Cleanup-bot mining.** Real errors in their original sentences are mined from the same bot’s edit history, keeping only fixes whose corrected word is in the Royal Institute Dictionary or PyThaiNLP word list and whose wrong and correct forms align cleanly, following the revision-mining tradition (Tanaka et al., 2020; Grundkiewicz and Junczys-Dowmunt, 2014), with the bot’s fix as gold. These items form a separate natural-error evaluation set (Section 5.5).

Registers. The clean text of the evaluated tiers spans news (ThaiSum), encyclopedic text (Thai Wikipedia), and Thai government press releases, all permissively redistributable.

3.4 Contamination control

Systems are evaluated on **corrupted passages that do not appear verbatim in the source corpora**. A residual risk remains: a model that memorized a clean source passage could reproduce it instead of finding and fixing the injected errors. We therefore probe memorization directly. Given the first half of 30 sentences per source, Typhoon 2.5 (30B) continues them verbatim for only 0.3% to 1.7% of the true second half on average, and never for half of it. We fill a separate held-out test split first from the least-memorized sources (details in Appendix C).

3.5 Length tiers

The same pipeline produces three tiers that vary passage length and error sparsity. **T1** (sentence) has 795 items with a mean length of 101 characters (95% of items fall between 43 and 174); **T2** (paragraph) has 224 items with a mean of 425 (270 to 577); and **T3** (page) has 122 items with a mean of 1,242 (963 to 1,530). They contain 974, 278, and 193 gold edits respectively. Injection is stratified by class quota rather than uniform: the class mix follows the skew of the attested pairs (the per-class share of T1 gold edits correlates with that of the attested pairs at Pearson $r = 0.80$) and is partially flattened so that every class has at least 115 gold edits at T1 (Appendix C). The error budget is **dense at the sentence level**, with roughly one error per 100 characters, and deliberately **sparse at paragraph and page length**, where most sentences are clean distractors. T2 and T3 therefore measure **precision under long correct context** (over-editing).

3.6 Quality control

Construction makes the correction of each injected error exact, but it cannot rule out two other problems: typos already present in the source text, and words with two accepted spellings. We add one check for each.

- **Cross-model consensus audit.** A typo left in the source text stays in the gold, so a model that fixes it is wrongly scored as over-editing. We flag an item when at least $k=3$ systems, and at least as many as reproduce the gold, return the *same* output that differs from the gold. We keep a flagged item when that output is the unchanged input or makes only some of the gold corrections (a *shared miss*), or differs only in punctuation or spacing (a *shared over-edit*). In T2 and T3 we drop every other

flagged item as a likely misspelling left in the *source*, which also drops a few items where the systems substitute an accepted variant, a formal word, or a standard loanword spelling; T1’s flagged items were judged individually. The audit removed 24 of 819 (T1), 26 of 250 (T2), and 28 of 150 (T3) items; Appendix B describes how flagged items were checked.

- **Variant reconciliation.** Before scoring, we map words with two accepted spellings (four groups, eight word forms; Appendix B) to the gold’s form. No injected wrong form belongs to the map, so the map cannot turn a missed error into a fix; it changes 2.35% of non-empty model outputs.

3.7 Human validation of the labels

The mechanism classes are assigned automatically. To measure how reliably the taxonomy can be applied, two native Thai speakers independently labeled 60 items (40 curated mined items and 20 synthetic ones), blind to the automatic label. Both judged all 60 marked spans to be genuine errors, and their class agreement is Krippendorff’s $\alpha = 0.752$ (Krippendorff, 2011). Where the raters agree (48 items), their class matches the automatic label on 35; most disagreements fall at the loanword-absorption and class-2 conventions (Appendix E).

4 Evaluation Protocol

4.1 Metrics

We grade at the TCC level with ERRANT-style span matching (Bryant et al., 2017). The **primary metric is correction F1**. We derive minimal TCC edits by aligning the wrong text with *both* the gold and the system output under the **same** Levenshtein alignment, so an output identical to the gold scores exactly 1.0, as in SIGHAN CSC scoring (Tseng et al., 2015). **Detection F1** credits the correct error site and ignores the replacement. **Clean-sentence fidelity** is the fraction of error-free sentences left unchanged, a precision signal for T2 and T3. The editor (Section 4.2) also gives **span-F0.5**, overall and per mechanism. We report 95% item-bootstrap confidence intervals (Appendix D) and test differences between systems with a **paired item-level bootstrap** (Koehn, 2004) over 10,000 resamples.

4.2 ERRANT-style edit-typer

We adapt the ERRANT/ChERRANT design (Bryant et al., 2017; Zhang et al., 2022) to Thai

orthography. Given a (wrong, correct) pair, the typer works in three steps. First, it extracts edits at the **TCC level**, with the same alignment the grader uses. Second, an edit often covers only part of a word, such as a single tone mark, so the typer widens each edit to the whole word that contains it, found with the dictionary-based word segmenter of PyThaiNLP (newmm; Phatthiyaphaibun et al., 2023). Third, a rule-based classifier compares the wrong and correct forms of that word character by character and assigns the class from what changed; for example, a changed tone mark gives class 1. If the corrected word is in a loan lexicon, the edit is class 6 (Appendix A gives the full rules). The same classifier labels the confusion set, and the typer also tags the edit operation (substitution, insertion, deletion, transposition, or mixed). Because it types both the gold and the system edits, it gives **span-F0.5 per mechanism** on any (wrong, correct) pair, without construction labels. On the dev split, it recovers 98.9% of the gold edit spans and agrees with the construction class on 98.1% of them. Because the typer reuses the construction-time classifier, we treat it as a **scoring instrument** for this benchmark, not a general Thai grammatical error classifier.

4.3 Models and inference

We evaluate 16 models, each in a single fixed configuration, at temperature 0 where supported, with one fixed Thai instruction prompt. The **frontier proprietary** models are GPT-5.5 (medium and high reasoning), Gemini 3.5 Flash, Claude Opus 4.7 (non-thinking), Claude Sonnet 4.6 (non-thinking), and GPT-5.4-mini. The **open-weight** models are Gemma-4-31B, DeepSeek V4 Pro and Flash, GLM-5.1, MiniMax-M2.7, and Qwen3.6-35B-A3B. The **Thai-adapted** models, which adapt general base models to Thai, are Typhoon 2.5 (30B) and three ThaiLLM 8B-instruct models served through the national ThaiLLM gateway, namely OpenThaiGPT (v7.2), Typhoon-S, and THaLLE (Pipatanakul et al., 2024; Pipatanakul and Taveekitworachai, 2026). Appendix G gives the prompts and a prompt-sensitivity ablation.

5 Results and Analysis

5.1 Leaderboard

Table 1 reports correction F1 on the three tiers and on the natural-error set (post-audit $n = 795/224/122/87$); the full table is in Appendix D. The frontier clusters at 0.95 to 0.97 on T1, and

model	correction F1				fidelity
	T1	T2	T3	Nat.	T3
GPT-5.5 (high)	0.973	0.968	0.862	0.907	0.977
GPT-5.5 (med)	0.969	0.954	0.866	0.901	0.977
Claude Opus 4.7	0.968	0.948	0.853	0.882	0.976
Gemini 3.5 Flash	0.965	0.960	0.825	0.850	0.963
Gemma-4-31B	0.950	0.954	0.898	0.926	0.984
GPT-5.4-mini	0.949	0.915	0.865	0.919	0.990
Claude Sonnet 4.6	0.947	0.905	0.825	0.877	0.974
DeepSeek V4 Flash	0.947	0.945	0.840	0.851	0.971
DeepSeek V4 Pro	0.926	0.926	0.754	0.821	0.958
GLM-5.1	0.846	0.811	0.657	0.775	0.954
Typhoon-S 8B [†]	0.782	0.702	0.617	0.703	0.964
THaLLE 8B [†]	0.749	0.594	0.507	0.742	0.814
Typhoon 2.5 30B [†]	0.742	0.632	0.476	0.741	0.899
OpenThaiGPT v7.2 [†]	0.741	0.476	0.463	0.725	0.901
MiniMax-M2.7	0.740	0.720	0.471	0.720	0.970
Qwen3.6-35B-A3B	0.705	0.384	0.310	0.588	0.838

Table 1: Correction F1 by tier and on the natural-error set (Nat.), and T3 clean-sentence fidelity (the fraction of error-free sentences left unchanged). **Bold** = column best; underlined = correction F1 tied with the best (paired bootstrap, $p \geq 0.05$); because the test is paired, a lower score can be tied while a higher one is not. [†] = Thai-adapted.

the confidence intervals separate three strata: the frontier and the strongest open-weight models, then GLM-5.1, then the Thai-adapted models, MiniMax, and Qwen. Within the frontier, T1 differences are mostly not significant: GPT-5.5 (high) is tied with GPT-5.5 (med) and Claude Opus 4.7.

General frontier models dominate, while the Thai-adapted models trail (0.74 to 0.78 on T1 and 0.46 to 0.62 on T3). The strongest **open-weight** model is a general 31B model, Gemma-4. The frontier leads it on T1 ($\Delta = 0.022$ against GPT-5.5 high, $p < 0.001$), and on T3 it has the highest point estimate (0.898) but is statistically tied with the frontier (against GPT-5.5 med, $\Delta = +0.032$, 95% CI $[-0.013, 0.077]$). Under this setup the evaluated Thai-adapted systems do not lead, and the gap widens with length (Gemma-4 against Typhoon 2.5 on T3: $\Delta = 0.422$, $p < 0.001$). This comparison is confounded by differences in base model, model scale, serving stack, and inference configuration, so it is not evidence about Thai adaptation in general.

5.2 Where models fail: finding versus fixing

We decompose every gold edit into **fixed**, **miscorrected** (the system edits exactly the gold span but with a different replacement), and **missed** (no system edit matches the gold span). Pooled over all 16 models, 0.866 of T1 gold edits are fixed, only 0.008

are miscorrected, and 0.126 are missed. The shares are 0.821, 0.005, and 0.173 at T3 and 0.815, 0.006, and 0.178 on natural errors (Section 5.5), and no single model’s miscorrected share exceeds 0.036 on T1 (Table 7, Appendix D). The pattern holds within every mechanism class, including the hardest (loanword: miscorrected 0.003 against missed 0.189). A model that rewrites a wider region around an error is scored as missed, but such edits are only 1.3 to 1.8% of gold edits per tier, so untouched errors still outnumber wrong fixes at least five to one (four to one without empty replies; Limitations). Under this benchmark’s conditions (formal register, single-reference gold, and one to three errors per passage), models rarely fail by choosing the wrong spelling once they locate an error, so what the leaderboard separates is **detection**. Spelling knowledge still matters, since recognizing a misspelling requires knowing the correct form; whether the finding holds for dense, interacting errors is open (Limitations).

5.3 Length robustness

Sentence-tier scores largely predict the page-tier ranking (Spearman $\rho = 0.93$), but the **drop from T1 to T3** differs widely between models (Figure 1a). The frontier loses only 0.08 to 0.14 correction F1 (GPT-5.5-med 0.969→0.866) and Gemma-4 only 0.05 (0.950→0.898), while Typhoon 2.5 and Qwen lose 0.27 and 0.40 (0.742→0.476 and 0.705→0.310). Measured per 10,000 clean TCCs, the over-edit rate is lower at T3 than at T1 for all 16 models (Figure 1b), although it rises from T2 to T3 for 11 of them, so over-edits per passage grow mainly with the amount of correct text. What separates models is the *level* of that rate: Typhoon 2.5, OpenThaiGPT, and Qwen over-edit at 3 to 35 times the rate of GPT-5.5 (med) at every tier (Appendix D). Clean-sentence **fidelity** agrees: the frontier holds at or above 0.95, while the heaviest over-editors sit at 0.77 to 0.90. These results suggest that **over-editing and missing reflect the same weak discrimination**. At T3, even the frontier models are limited more by over-editing than by detection: they fix 0.94 to 0.98 of gold edits, but their precision is 0.71 to 0.80 (GPT-5.4-mini, with recall 0.85 and precision 0.88, is the exception).

5.4 Per-mechanism difficulty

The edit-typer types the system’s edits as well as the gold’s, so it separates recall from precision for each mechanism (Table 2). Mean recall is lowest for loanwords at T1 (0.81 against 0.86 to 0.90), and the

class	T1			T3		
	R	P	F0.5	R	P	F0.5
1 tone mark	0.898	0.902	0.900	0.801	0.687	0.692
2 consonant-pair	0.876	0.960	0.940	0.812	0.954	0.915
3 silent-consonant	0.886	0.936	0.923	0.800	0.770	0.766
4 vowel	0.857	0.956	0.929	0.831	0.880	0.861
5 cluster	0.863	0.822	0.826	0.843	0.656	0.677
6 loanword	0.808	0.989	0.926	0.819	0.941	0.899

Table 2: Per-mechanism span recall (R), precision (P), and span-F0.5 from the edit-typer: the mean of the 16 models’ per-class values, each model weighted equally. A system edit that matches no gold edit lowers the precision of the class the typer assigns to it; edits that fit no class, such as spacing or rewording (6% to 84% of a model’s unsupported edits), enter no column.

construction labels agree (Table 8, Appendix D); the failure is again missed rather than miscorrected. This mean hides a split: the nine systems of the top stratum (Section 5.1) find 0.94 to 1.00 of loanword errors, while the four Thai-adapted models and MiniMax find 0.32 to 0.70. Precision shows where models change text that needs no change, through *unsupported* edits that match no gold edit. Class 5 (reordered, doubled, or inserted letters) has the lowest precision for 15 of the 16 models at T1, and at T3 class 5 or tone marks are lowest for 14 of them. Loanword precision is high by construction: the typer assigns class 6 only when the output word is in its loan lexicon, so a change that turns a correct loanword into another form is usually charged to another class.

5.5 Natural errors

We evaluate all 16 systems on the mined cleanup-bot items, each a real error in its original encyclopedic sentence. The consensus audit of Section 3.6 removes 16 of the 103 items, where the agreeing systems correct more than the recorded bot fix does (Appendix F). On the remaining 87 items, mean correction F1 is 0.81 against 0.87 on T1, and the system ranking transfers (Spearman $\rho = 0.88$ with T1; Table 1). Without the audit, mean F1 is 0.72, mostly because of those 16 items, and ρ is again 0.88.

6 Conclusion

We presented ThaiEditBench, a mechanism-typed, length-stratified benchmark for Thai spelling correction. Across 16 models, **detection** rather than the choice of spelling separates systems, and weaker models accumulate over-edits in long text.

Limitations

Sparse error density (primary). The long tiers (T2 and T3) are deliberately sparse, with 1 to 3 errors scattered across otherwise-clean passages, to isolate precision under long correct context. We therefore do **not** test **high error-density** regimes such as heavy-typo input, degraded OCR, or L2 writing, where errors co-occur, interact, mask one another, and correction order matters. Extending ThaiEditBench to dense multi-error passages, and testing whether the detection finding survives when errors crowd each other, is the most important next step.

A fixed inventory of attested errors. Every injected error comes from 1,044 attested pairs, each mapping one wrong form to one correct form. The benchmark therefore does not cover misspellings outside the inventory, and a system that consulted the released inventory would be answering a different, tool-augmented question; all systems are evaluated zero-shot without it. The class mix follows the distribution of attested misspelling types, not the frequency of errors in running text.

Scope of the natural-error check. The audited natural set has 87 items from one cleanup bot in one register, with only 7 loanword and 5 class-5 edits, so it supports the ranking and the decomposition but not per-class conclusions; loanword behaviour is measured mainly on injected errors. The audit relies on cross-model consensus, and the bot’s fix is taken as gold, so the natural gold is less certain than the construction gold of the tiers. A larger, naturally distributed test set, which the released mining pipeline supports, is the main way to strengthen this check.

Label reliability. The annotation study covers 60 items and two raters. Per-class agreement on ten items per class is noisy (α from 0.49 to 1.00) and lowest for class 6, whose boundary depends on the absorption convention, and the automatic class matches the rater consensus on 35 of 48 items. Per-mechanism results near the class 6 and class 2 boundaries should be read with this uncertainty.

Configured-system comparison. Each system is evaluated under one fixed configuration and one fixed Thai prompt. Reasoning modes, token budgets, provider defaults, serving stacks, and model scale are not controlled, so the leaderboard reflects configured-system behaviour rather than an intrinsic ranking of base-model capability. A prompt ablation on two systems shifts correction F1 by at

most 0.07 without reordering them (Appendix G), but we do not sweep prompts across all systems. A reply that exhausts its token budget is empty and scored as the unchanged input, so its errors count as missed. This affects 112 of 795 T1 items for GLM-5.1, 93 for MiniMax, and 58 for Qwen, and 51 of 122 T3 items for MiniMax; empty replies account for 17% of missed edits at T1 and 29% at T3. For these systems, a low score partly reflects failure to deliver an answer rather than failure to detect errors, and excluding empty replies, untouched errors outnumber wrong fixes four to one rather than five to one.

Acknowledgments

We thank Pachaya Sailamul for help with the annotation study.

References

- Riadh Belkebir and Nizar Habash. 2021. [Automatic error type annotation for Arabic](#). In *Proceedings of the 25th Conference on Computational Natural Language Learning*, pages 596–606, Online. Association for Computational Linguistics.
- Christopher Bryant, Mariano Felice, Øistein E. Andersen, and Ted Briscoe. 2019. [The BEA-2019 shared task on grammatical error correction](#). In *Proceedings of the Fourteenth Workshop on Innovative Use of NLP for Building Educational Applications*, pages 52–75, Florence, Italy. Association for Computational Linguistics.
- Christopher Bryant, Mariano Felice, and Ted Briscoe. 2017. [Automatic annotation and evaluation of error types for grammatical error correction](#). In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 793–805, Vancouver, Canada. Association for Computational Linguistics.
- Daniel Dahlmeier and Hwee Tou Ng. 2012. [Better evaluation for grammatical error correction](#). In *Proceedings of the 2012 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 568–572, Montréal, Canada. Association for Computational Linguistics.
- Roman Grundkiewicz and Marcin Junczys-Dowmunt. 2014. [The WikEd error corpus: A corpus of corrective Wikipedia edits and its application to grammatical error correction](#). In Adam Przepiórkowski and Maciej Ogrodniczuk, editors, *Advances in Natural Language Processing – Lecture Notes in Computer Science, Vol. 8686 (PolTAL 2014)*, pages 478–490. Springer.

- Philipp Koehn. 2004. [Statistical significance tests for machine translation evaluation](#). In *Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing*, pages 388–395, Barcelona, Spain. Association for Computational Linguistics.
- Klaus Krippendorff. 2011. [Computing Krippendorff’s alpha-reliability](#). Departmental Papers (ASC) 43, Annenberg School for Communication, University of Pennsylvania.
- Anuruth Lertpiya, Tawunrat Chalothorn, and Pakpoom Buabthong. 2023. [How to progressively build Thai spelling correction systems?](#) *IEEE Access*, 11:72704–72716.
- Anuruth Lertpiya, Tawunrat Chalothorn, and Ekapol Chuangsuwanich. 2020. [Thai spelling correction and word normalization on social text using a two-stage pipeline with neural contextual attention](#). *IEEE Access*, 8:133403–133419.
- Peerat Limkonchotiwat, Wannaphong Phatthiyaphaibun, Raheem Sarwar, Ekapol Chuangsuwanich, and Sarana Nutanong. 2021. [Handling cross- and out-of-domain samples in Thai word segmentation](#). In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 1003–1016, Online. Association for Computational Linguistics.
- Chao-Lin Liu, Min-Hua Lai, Kan-Wen Tien, Yi-Hsuan Chuang, Shih-Hung Wu, and Chia-Ying Lee. 2011. [Visually and phonologically similar characters in incorrect Chinese words: Analyses, identification, and applications](#). *ACM Transactions on Asian Language Information Processing*, 10(2):10:1–10:39.
- Pakawat Nakwijit and Matthew Purver. 2022. [Misspelling semantics in Thai](#). In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 227–236, Marseille, France. European Language Resources Association.
- Courtney Napoles, Keisuke Sakaguchi, Matt Post, and Joel Tetreault. 2015. [Ground truth for grammatical error correction metrics](#). In *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, pages 588–593, Beijing, China. Association for Computational Linguistics.
- Courtney Napoles, Keisuke Sakaguchi, and Joel Tetreault. 2017. [JFLEG: A fluency corpus and benchmark for grammatical error correction](#). In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers*, pages 229–234, Valencia, Spain. Association for Computational Linguistics.
- Wannaphong Phatthiyaphaibun, Korakot Chaovavanich, Charin Polpanumas, Arthit Suriyawongkul, Lalita Lowphansirikul, Pattarawat Chormai, Peerat Limkonchotiwat, Thanathip Suntornrati, and Can Udomcharoenchaikit. 2023. [PyThaiNLP: Thai natural language processing in python](#). In *Proceedings of the 3rd Workshop for Natural Language Processing Open Source Software (NLP-OSS 2023)*, pages 25–36, Singapore. Association for Computational Linguistics.
- Kunat Pipatanakul, Potsawee Manakul, Natapong Nitarach, Warit Sirichotedumrong, Surapon Nonesung, Teetouch Jaknamon, Parinthapat Pengpun, Pittawat Taveekitworachai, Adisai Na-Thalang, Sittipong Sripaisarnmongkol, Krisanapong Jirayoot, and Kasima Tharnpipitchai. 2024. [Typhoon 2: A family of open text and multimodal Thai large language models](#). *Preprint*, arXiv:2412.13702.
- Kunat Pipatanakul and Pittawat Taveekitworachai. 2026. [Typhoon-S: Minimal open post-training for sovereign large language models](#). *Preprint*, arXiv:2601.18129.
- Royal Institute. 2013. *Photchananukrom Chabap Ratchabandittayasathan Pho. So. 2554 [The Royal Institute Dictionary, B.E. 2554]*. Nanmeebooks Publications, Bangkok. In Thai. ISBN 9786167073569.
- Yu Tanaka, Yugo Murawaki, Daisuke Kawahara, and Sadao Kurohashi. 2020. [Building a Japanese typo dataset from Wikipedia’s revision history](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics: Student Research Workshop*, pages 230–236, Online. Association for Computational Linguistics.
- Thanaruk Theeramunkong, Virach Sornlertlamvanich, Thanasan Tanhermhong, and Wirat Chinnan. 2000. [Character cluster based Thai information retrieval](#). In *Proceedings of the Fifth International Workshop on Information Retrieval with Asian Languages (IRAL 2000)*, pages 75–80, Hong Kong, China. Association for Computing Machinery.
- Yuen-Hsien Tseng, Lung-Hao Lee, Li-Ping Chang, and Hsin-Hsi Chen. 2015. [Introduction to SIGHAN 2015 bake-off for Chinese spelling check](#). In *Proceedings of the Eighth SIGHAN Workshop on Chinese Language Processing*, pages 32–37, Beijing, China. Association for Computational Linguistics.
- Shih-Hung Wu, Chao-Lin Liu, and Lung-Hao Lee. 2013. [Chinese spelling check evaluation at SIGHAN bake-off 2013](#). In *Proceedings of the Seventh SIGHAN Workshop on Chinese Language Processing*, pages 35–42, Nagoya, Japan. Asian Federation of Natural Language Processing.
- Soyoung Yoon, Sungjoon Park, Gyuwan Kim, Junhee Cho, Kihyo Park, Gyu Tae Kim, Minjoon Seo, and Alice Oh. 2023. [Towards standardizing Korean grammatical error correction: Datasets and annotation](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6713–6742, Toronto, Canada. Association for Computational Linguistics.
- Liang-Chih Yu, Lung-Hao Lee, Yuen-Hsien Tseng, and Hsin-Hsi Chen. 2014. [Overview of SIGHAN 2014 bake-off for Chinese spelling check](#). In *Proceedings of the Third CIPS-SIGHAN Joint Conference on Chinese Language Processing*, pages 126–132, Wuhan, China. Association for Computational Linguistics.

Zheng Yuan and Christopher Bryant. 2021. [Document-level grammatical error correction](#). In *Proceedings of the 16th Workshop on Innovative Use of NLP for Building Educational Applications*, pages 75–84, Online. Association for Computational Linguistics.

Yue Zhang, Zhenghua Li, Zuyi Bao, Jiacheng Li, Bo Zhang, Chen Li, Fei Huang, and Min Zhang. 2022. [MuCGEC: a multi-reference multi-source evaluation dataset for Chinese grammatical error correction](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3118–3130, Seattle, United States. Association for Computational Linguistics.

A Mechanism taxonomy in detail

Each class is defined by the orthographic *mechanism* of the fault, not its surface character. Multiple worked examples per class:

- **1 tone mark:** ค่ะ→คะ, นะ→นะ, จั๊กจั่น→จ๊กจั่น, กระทิง→กระต๊ิง
- **2 consonant-pair:** กฎหมาย→กฏหมาย, กบฏ→กบฏ, อากาศ→อากาศ
- **3 silent-consonant:** กลยุทธ์→กลยุทธ์, ขาดดุลย์→ขาดดุล, เทห์→เท
- **4 vowel:** สังเกต→สังเกต, มาตราฐาน→มาตรฐาน, ณะเพาะ→เนเพาะ
- **5 cluster/metathesis:** กระทั่งหัน→กะทั่งหัน, กระบ้งลม→กะบ้งลม
- **6 loanword:** กราฟฟิค→กราฟิก, กองสุล→กองสุล, ลิ้งค์→ลิงก์

Class 2 scope. Class 2 covers swaps within a homophone group that contains an Indic-origin letter, for example ศ/ษ/ส, ฑ/ฒ, and น/ณ, plus the final-stop pair บ/ป. Swaps between letters outside these groups, such as ฟ/ฟ, fall in class 5 together with other structural edits.

Loanword-absorption rule. Only *contemporary* foreign loans, chiefly modern English technical and commercial terms (กราฟิก, ดิจิทัล, ลิงก์), are class 6. Long-integrated borrowings are treated as **native** even though etymologically foreign: Portuguese (กะละแม), Burmese (กะปี้), Mon (กะพง), Khmer (กบฏ), and Tamil, Persian, and Sanskrit–Pali doublets. Borderline cases (e.g. กองสุล, *consul*) are resolved by the Royal Institute Dictionary listing + Wiktionary borrowed-term categories, and the tag is recorded per item so the rule is auditable.

Automatic class assignment. The rule-based classifier used to label the confusion set and inside

the edit-typing (Section 4.2) compares the wrong and correct forms of a word character by character. It checks, in order, for a swap within one of the homophone consonant groups above (class 2), an added or removed silent-consonant mark or a change in the consonant it silences (class 3), reordered, doubled, or inserted letters (class 5), a changed tone mark (class 1), and a changed vowel sign (class 4), and it assigns the first class that applies. Any other consonant change falls in class 5. If the corrected word is in the loan lexicon, which holds the correct forms of the class-6 confusion pairs and a few proper nouns, the class is 6 whatever the character change.

B Gold quality control and the consensus audit

Why construction-gold still needs a cleanup pass.

Injection guarantees that applying the gold edits recovers the clean original (verified 100%), so the *injected* error’s correction is exact. It does **not** guarantee that the **source passage was typo-free**: the permissive corpora contain a small rate of residual orthographic errors. When an item’s source carries such a residual typo, the gold inherits it, so a model that fixes *that* genuinely wrong span is unfairly scored as a miss or an over-edit. We find and remove such items with a cross-model consensus audit.

Cross-model consensus audit. For every item we pool the system outputs (variant-reconciled) and look for a *consensus* output: the same normalized string from at least $k=3$ systems, and from at least as many systems as reproduce the gold, that differs from the gold. Each flagged item falls into one of three cases. (i) If the consensus equals the corrupted input, or makes some of the gold corrections and no other change, this is a *shared miss*, the gold is correct, and we **keep** the item. (ii) If the consensus differs only in punctuation or whitespace, or, in the judged T1 check, is a register or spelling variant, this is a systematic *over-edit*, the precision signal we want, and we **keep** it. (iii) If the consensus is a *different* spelling fix at an unmarked span, this is a residual *source* typo, and we **drop** it. Only case (iii) is a gold defect, and the explicit test of whether the consensus equals the input prevents conflating it with shared misses and over-pruning. The audit removed 24 (T1), 26 (T2), and 28 (T3) items, for example ทั้งสิ้น→ทั้งสิ้น, ปฏิกษะ→ปฏิกษะ, and โดบ→โดย, all residual *source* typos. **How flagged**

items were checked. For T2 and T3 the check is automatic and uses the outputs of all 16 systems: after variant reconciliation, which maps the informal pronoun `เค้า` to `เขา`, a flagged item is dropped unless the consensus equals the input, is a partial fix on the way to the gold, or differs from the gold only in punctuation or whitespace. The rule cannot tell a source typo from an accepted variant or register change outside the variant map, so it also drops a few such items, for example `วิกฤติ`→`วิกฤต`, the standard loanword spelling `เซ็ด`→`เซต`, and the informal words `พวกเค้า` and `เท่าไหร่`. The T1 audit pooled 9 of the 16 systems (GPT-5.5 med and high, GPT-5.4-mini, Gemini 3.5 Flash, Claude Sonnet 4.6, DeepSeek V4 Pro and Flash, GLM-5.1, and Typhoon 2.5), and its flagged items were judged individually rather than filtered by rule. The audit flagged 70 items. Six differed from the gold only in punctuation and were kept; we judged the other 64 against the three cases with LLM assistance, dropping 23 as residual source typos and keeping 41, among them 14 informal-to-formal rewritings such as `มัย`→`ไหม` that the minimal-edit contract treats as over-edits. One further item was dropped because its source sentence itself lacks a syllable (`ความใจ` for `ความเข้าใจ`). Judged and rule-based checks differ at the margin: run on T1 with all 16 systems, the automatic rule would drop 21 further items. One T1 item and one T2 item contain an error that no model fixes; both inject `แน่นหนา` for `หนาแน่น`, two valid words, so only the context marks the error. We release the audit reports, which list the gold-versus-consensus difference for every flagged item.

Variant map. The four equivalence groups are an orthographic doublet (`สิทธิ/สิทธิ์`), two loanword transliterations (`ดิจิทัล/ดิจิทัล` and `ออร์แกนิก/ออร์แกนิก`), and an informal-to-formal pronoun pair (`เค้า/เขา`). Words in which the final mark is mandatory, such as `ลิขสิทธิ์`, are protected from reconciliation. Beyond these groups, orthographic correction is essentially single-reference, since a word has one correct spelling, so we do not adopt multi-reference GEC scoring.

C Construction details

Confusion set. Wrong→correct pairs were harvested from the Thai Wiktionary appendix of commonly misspelled words and from the fix history of a Thai Wikipedia spelling-cleanup bot, then deduplicated and cleaned (fragments, multi-token and spacing-only pairs, and context-dependent real-

word pairs removed; one such pair, `แน่นหนา/หนาแน่น`, was not caught), giving 1,044 canonical pairs. A rule-based classifier assigns the mechanism class from the character difference; it agrees with 90 hand-labeled bot-fix pairs on 96.5%. Etymology is assigned from the Wiktionary borrowed-term categories under the absorption rule, which moves contemporary loans into class 6. We check canonical forms against the Royal Institute Dictionary word list distributed with PyThaiNLP: 87.5% are listed, and the unlisted forms include modern loanwords and common words that the list omits, such as `นะคะ`. Each pair is also tagged with its edit operation. Table 4 gives the per-class sizes.

Injection and mining. Injection uses **whole-token-run matching** so the 51 valid short forms (≤ 3 characters) that coincide with a confusion target are never corrupted. Cleanup-bot mining reads the bot’s contribution history on Thai Wikipedia and keeps a fix only when the corrected word is in the Royal Institute Dictionary word list or PyThaiNLP’s general Thai word list and the wrong and correct forms align without shifting neighbouring characters; the mined items are kept out of the three tiers. The curated subset used in the annotation study keeps at most two contexts per distinct error and drops fragments of two characters or fewer.

Memorization probe. For each of the four sources we sample 30 clean sentences of 90 to 180 characters, give Typhoon 2.5 (30B) the first half at temperature 0 with the instruction `เติมข้อความต่อไปนี้ให้สมบูรณ์โดยใช้ถ้อยคำเดิมที่ถูกต้องที่สุด ตอบเฉพาะส่วนที่เติมต่อ ห้ามอธิบาย` (‘Complete the following text with its exact original wording. Reply with the continuation only, without explanation.’), and allow up to 120 output tokens. The score is the longest common prefix between the continuation and the true second half, ignoring whitespace, divided by the length of the true second half. The mean score is 0.003 (news), 0.004 (government press releases), 0.012 (Wikipedia), and 0.017 (Thai Constitution text, used only in the train split), and no sentence scores 0.5 or more. The probe uses a single model, so it does not measure memorization by the other systems directly. Items that share a clean sentence are then kept together as one group, and groups are assigned to splits class by class (by the class of the group’s first item): within each class, groups are shuffled with a fixed seed and ordered by their source’s mean score, and the lowest-scoring 34%

tier	unit	items	chars	err/it	news/gov/enc
T1 sent.	1 sent.	795	101	1.23	275/368/152
T2 para.	≈5 sent.	224	425	1.24	80/79/65
T3 page	≈15 sent.	122	1,242	1.58	38/43/41

Table 3: Per-tier statistics (post-audit; Section 3.6): mean characters per item and gold edits per item. The tiers are 100% confusion-set injection; the cleanup-bot items form the separate natural-error set.

class	pairs	T1	T2	T3
1 tone mark	43	163	42	32
2 consonant-pair	168	137	37	26
3 silent-consonant	177	166	54	26
4 vowel	189	182	60	38
5 cluster/metathesis	338	211	55	54
6 loanword	129	115	30	17
total	1,044	974	278	193

Table 4: Confusion-set pairs and injected gold edits per class and tier; the grader scores the TCC-aligned edits, which number 982, 278, and 195 at T1–T3. The class shares of T1 gold edits correlate with those of the attested pairs at Pearson $r = 0.80$ (Spearman 0.82); cluster/metathesis is the largest class in both.

(mined groups first) form the held-out test split, followed by the public test (32%), dev (24%), and train splits.

Release and license. The injected items are released as train (249), dev (606), and public-test (819 before the audit) splits, plus a held-out test split (868) drawn from the least-memorized sources; the results in this paper use the audited public-test split as T1. We release the benchmark for research use under CC BY-SA (data) and the MIT license (code), consistent with the permissive redistribution terms of the source corpora.

D Full results

Pooled per-class outcome split (T1, all 16 models).

class	gold edits	fixed	miscorrected	missed
1 tone	2,608	0.896	0.011	0.092
2 consonant	2,192	0.876	0.006	0.117
3 silent-consonant	2,640	0.887	0.010	0.103
4 vowel	2,896	0.857	0.002	0.140
5 cluster	3,392	0.863	0.012	0.126
6 loanword	1,968	0.808	0.003	0.189

Over-edits per 100 items across tiers (absolute accumulation).

model	T1	T2	T3
GPT-5.5 (med)	7	8	38
Gemma-4-31B	6	8	30
GLM-5.1	10	23	108
Typhoon 2.5 30B	32	81	192
OpenThaiGPT v7.2	20	110	197
Qwen3.6-35B-A3B	59	268	479

Over-edit rate per 10,000 clean TCCs (length-normalized; the Figure 1b data). The per-unit rate is lower at T3 than at T1 for every model (though higher at T3 than at T2 for most), so the per-item growth above mainly reflects the amount of correct text.

model	T1	T2	T3
GPT-5.5 (med)	11.7	3.0	5.1
Gemma-4-31B	10.9	3.0	4.0
GLM-5.1	17.0	9.2	14.5
Typhoon 2.5 30B	54.3	32.1	25.7
OpenThaiGPT v7.2	33.5	43.3	26.4
Qwen3.6-35B-A3B	100.8	105.7	64.2

Register effect (pooled correction recall, T1/T2/T3). encyclopedic 0.82/0.79/0.76; news 0.87/0.85/0.82; government 0.89/0.90/0.88. Per-model $\times$ class correction recall and span-F0.5 per type are released with the data.

E Annotation study

Design. The 60 items comprise the 40 curated mined items (Appendix C) and 20 synthetic items chosen so that each automatic class has ten items. Each item shows one marked span with its proposed correction. Raters answer two questions per item: whether the marked span is a genuine orthographic error, and which of the six classes it belongs to, following a written guideline with the class definitions and the two conventions of Appendix A. No automatic label is shown. The two raters, one of them an author, worked independently. Etymology and edit-operation tags are not annotated, since they are a dictionary lookup and a deterministic function of the edit.

Results. Error validity: both raters accepted all 60 spans. Class agreement: Krippendorff’s $\alpha = 0.752$ (nominal, $n = 60$). Table 9 gives one-vs-rest α and raw agreement per automatic class; with ten items per class these values are noisy.

Where labels differ. The raters split on 12 items and agree with each other against the automatic label on 13. Of these 25, 17 involve the loanword-absorption rule in one of two directions: raters labeled the surface mechanism of a contemporary loan (for example, a consonant swap in เวียดนาม→เวียดนาม rated class 2 rather than 6),

model	T1 det	T1 corr	T1 span-F0.5	T2 corr	T2 fidelity	T3 corr	T3 fidelity
GPT-5.5 (high)	0.973	0.973	0.961	0.968	0.990	0.862	0.977
GPT-5.5 (med)	0.969	0.969	0.956	0.954	0.983	0.866	0.977
Claude Opus 4.7	0.971	0.968	0.956	0.948	0.986	0.853	0.976
Gemini 3.5 Flash	0.969	0.965	0.951	0.960	0.983	0.825	0.963
Gemma-4-31B	0.954	0.950	0.947	0.954	0.988	0.898	0.984
GPT-5.4-mini	0.959	0.949	0.951	0.915	0.986	0.865	0.990
Claude Sonnet 4.6	0.952	0.947	0.938	0.905	0.977	0.825	0.974
DeepSeek V4 Flash	0.952	0.947	0.939	0.945	0.986	0.840	0.971
DeepSeek V4 Pro	0.928	0.926	0.910	0.926	0.970	0.754	0.958
GLM-5.1	0.851	0.846	0.879	0.811	0.961	0.657	0.954
Typhoon-S 8B [†]	0.788	0.782	0.841	0.702	0.961	0.617	0.964
THaLLE 8B [†]	0.758	0.749	0.787	0.594	0.774	0.507	0.814
Typhoon 2.5 30B [†]	0.776	0.742	0.728	0.632	0.881	0.476	0.899
OpenThaiGPT v7.2 [†]	0.755	0.741	0.776	0.476	0.887	0.463	0.901
MiniMax-M2.7	0.747	0.740	0.801	0.720	0.974	0.471	0.970
Qwen3.6-35B-A3B	0.721	0.705	0.652	0.384	0.776	0.310	0.838

Table 5: Full leaderboard: detection and correction F1, span-F0.5 from the edit-typer, and clean-sentence fidelity across the three length tiers. **Bold** = column best; [†] = Thai-adapted. Table 1 marks which systems are statistically tied with each column’s best. Table 6 gives the confidence intervals. Per-class recall and per-type span-F0.5 are released with the data.

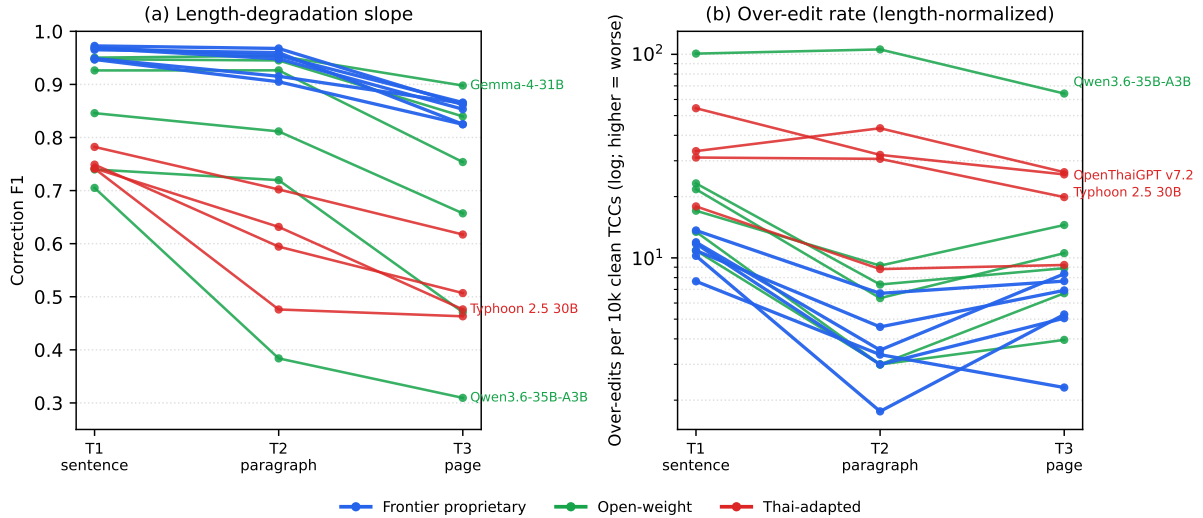


Figure 1: Length robustness. (a) Correction-F1 slope from T1 to T3. The frontier (blue) and the strongest open-weight model (Gemma-4, green) stay high while weak models slope off. (b) Over-edit rate per 10,000 clean TCCs on a log scale, where higher is worse. The rate is lower at T3 than at T1 for every model, so the per-page accumulation (the absolute counts in this appendix) mainly reflects the amount of correct text. Weak models over-edit at many times the frontier’s per-unit rate at every tier.

or recognized a loan that the loan lexicon missed (เขอรมันน์→เขอรมนน์, typed class 5 but rated 6); old borrowings such as เวทมนตร์ split the raters between class 3 and 6. Four more are homophone swaps outside the Indic-origin groups (คน→คน three times, and กบฏ→กบฏ), which the taxonomy places in class 5 and both raters placed in class 2. The remaining four concern whether a stray or misplaced diacritic counts as a tone-mark, vowel, or silent-consonant error. The disagreements therefore sit at the class boundaries fixed by convention

rather than at error perception.

F Natural-error evaluation

The natural set comes from the edit history of a Thai Wikipedia spelling-cleanup bot: 103 items, each a real error in its original encyclopedic sentence, with the bot’s fix as gold. We apply the consensus audit of Section 3.6 with the same rule as the tiers: at least $k = 3$ systems, and at least as many as reproduce the gold, must agree on an output that differs from the gold. The audit drops an item when that out-

model	T1	T2	T3
GPT-5.5 (high)	[0.964, 0.980]	[0.950, 0.983]	[0.809, 0.910]
GPT-5.5 (med)	[0.960, 0.977]	[0.933, 0.973]	[0.819, 0.911]
Claude Opus 4.7	[0.958, 0.978]	[0.924, 0.970]	[0.783, 0.918]
Gemini 3.5 Flash	[0.955, 0.975]	[0.942, 0.977]	[0.764, 0.883]
Gemma-4-31B	[0.938, 0.962]	[0.931, 0.974]	[0.859, 0.934]
GPT-5.4-mini	[0.936, 0.960]	[0.887, 0.942]	[0.822, 0.902]
Claude Sonnet 4.6	[0.933, 0.959]	[0.873, 0.935]	[0.754, 0.891]
DeepSeek V4 Flash	[0.935, 0.959]	[0.919, 0.968]	[0.788, 0.889]
DeepSeek V4 Pro	[0.912, 0.939]	[0.902, 0.950]	[0.692, 0.813]
GLM-5.1	[0.823, 0.867]	[0.767, 0.854]	[0.537, 0.773]
Typhoon-S 8B [†]	[0.759, 0.805]	[0.652, 0.750]	[0.549, 0.682]
THaLLE 8B [†]	[0.715, 0.780]	[0.540, 0.649]	[0.426, 0.589]
Typhoon 2.5 30B [†]	[0.716, 0.767]	[0.581, 0.684]	[0.406, 0.550]
OpenThaiGPT v7.2 [†]	[0.701, 0.777]	[0.385, 0.569]	[0.386, 0.544]
MiniMax-M2.7	[0.714, 0.765]	[0.672, 0.765]	[0.391, 0.550]
Qwen3.6-35B-A3B	[0.673, 0.736]	[0.312, 0.472]	[0.227, 0.421]

Table 6: 95% item-bootstrap confidence intervals on correction F1 (10,000 resamples). [†] = Thai-adapted.

model	fixed	miscorrected	missed
GPT-5.5 (high)	0.993	0.000	0.007
Gemma-4-31B	0.956	0.004	0.040
Qwen3.6-35B-A3B	0.817	0.018	0.165
Typhoon 2.5 30B	0.764	0.036	0.201
All 16, T1	0.866	0.008	0.126
All 16, T3	0.821	0.005	0.173
All 16, natural	0.815	0.006	0.178

Table 7: Gold-edit outcomes (T1 unless stated). The miscorrected share stays near zero throughout.

class	T1	T2	T3
1 tone mark	0.896	0.868	0.801
2 consonant-pair	0.876	0.885	0.812
3 silent-consonant	0.887	0.865	0.800
4 vowel	0.857	0.870	0.831
5 cluster/metathesis	0.863	0.807	0.843
6 loanword	0.808	0.790	0.819
all classes	0.866	0.849	0.821
spread (max-min)	0.088	0.096	0.042

Table 8: Fraction of gold edits fixed, by mechanism class and tier, pooled over all 16 models (construction labels). Table 4 gives the injected errors per class; the scored TCC edit spans per item can differ slightly after alignment.

put makes a spelling change the gold does not, and keeps it when the output equals the input, differs only in whitespace or punctuation, or is a partial fix on the way to the gold. It removes 16 items. In some, the consensus also fixes a second, unmarked error in the sentence (for example เพียง→เพ็ญ); in others, the recorded fix is itself incomplete and the consensus completes it (ต่ำ recorded as ต่ำ, น้ำ as น้ำ, and สร้าง as สร้าง, where the intended words

automatic class	α (one-vs-rest)	agreement
1 tone mark	0.793	1.00
2 consonant-pair	0.921	0.90
3 silent-consonant	0.611	0.60
4 vowel	0.773	0.70
5 cluster/metathesis	1.000	1.00
6 loanword	0.485	0.60
all (multiclass)	0.752	0.80

Table 9: Inter-rater agreement per automatic class (ten items each).

are ต่ำ, น้ำ, and สร้าง). Dropping an item whenever three systems agree, however many reproduce the gold, would remove 24 items (79 kept; mean correction F1 0.81, $\rho = 0.92$), including one where the systems change a correctly spelled film title. The audited set has 87 items with 87 recorded fixes (88 aligned gold edits). Classified without the loanword rule, the fixes fall in classes 1: 27, 2: 21, 3: 14, 4: 20, and 5: 5; the edit-typer, which applies the rule, types 7 of the 88 gold edits as loanwords. Table 10 gives per-model results on the audited set, with correction F1 and the miscorrected share on all 103 items for comparison. On the audited set the ranking correlates with the T1 ranking at Spearman $\rho = 0.88$ (Pearson $r = 0.93$); on all 103 items, also at $\rho = 0.88$ ($r = 0.93$).

G Prompt and inference

Prompt. All models receive the same Thai system message and instruction. The system message is คุณเป็นเครื่องมือตรวจแก้การสะกดคำภาษาไทย ตามอักขรวิธีมาตรฐานของราชบัณฑิตยสภา (‘You are a tool that checks and corrects Thai spelling

model	audited set ($n = 87$)						all items ($n = 103$)	
	corr F1	95% CI	det F1	fixed	miscorrected	missed	corr F1	miscorrected
Gemma-4-31B	0.926	[0.879, 0.966]	0.937	0.920	0.011	0.068	0.819	0.058
GPT-5.4-mini	<u>0.919</u>	[0.866, 0.966]	0.931	0.909	0.011	0.080	0.811	0.058
GPT-5.5 (high)	<u>0.907</u>	[0.868, 0.944]	0.907	0.943	0.000	0.057	0.800	0.048
GPT-5.5 (med)	<u>0.901</u>	[0.857, 0.943]	0.901	0.932	0.000	0.068	0.791	0.048
Claude Opus 4.7	<u>0.882</u>	[0.790, 0.954]	0.882	0.932	0.000	0.068	0.791	0.038
Claude Sonnet 4.6	<u>0.877</u>	[0.811, 0.934]	0.889	0.852	0.011	0.136	0.771	0.058
DeepSeek V4 Flash	0.851	[0.765, 0.922]	0.851	0.841	0.000	0.159	0.757	0.048
Gemini 3.5 Flash	<u>0.850</u>	[0.749, 0.944]	0.850	0.966	0.000	0.034	0.757	0.038
DeepSeek V4 Pro	0.821	[0.735, 0.904]	0.821	0.909	0.000	0.091	0.729	0.038
GLM-5.1	0.775	[0.686, 0.862]	0.775	0.761	0.000	0.239	0.689	0.048
THaLLE 8B [†]	0.742	[0.658, 0.818]	0.742	0.670	0.000	0.330	0.670	0.048
Typhoon 2.5 30B [†]	0.741	[0.653, 0.826]	0.751	0.795	0.011	0.193	0.655	0.048
OpenThaiGPT v7.2 [†]	0.725	[0.638, 0.807]	0.725	0.659	0.000	0.341	0.630	0.048
MiniMax-M2.7	0.720	[0.632, 0.797]	0.720	0.614	0.000	0.386	0.663	0.038
Typhoon-S 8B [†]	0.703	[0.610, 0.784]	0.717	0.580	0.011	0.409	0.626	0.038
Qwen3.6-35B-A3B	0.588	[0.494, 0.680]	0.623	0.761	0.045	0.193	0.544	0.067

Table 10: Natural-error evaluation. **Bold** = best on the audited set; underlined = not separable from the best by a paired bootstrap test ($p \geq 0.05$); [†] = Thai-adapted.

according to the standard orthography of the Royal Society’). The T1 instruction, with the input in place of {text}, is:

จงแก้คำที่สะกดผิดในประโยคต่อไปนี้ให้ถูกต้องตามอักขรวิธี แก้เฉพาะตัวพยัญชนะ/สระ/วรรณยุกต์/ตัวการันต์ที่สะกดผิดเท่านั้น ห้ามเปลี่ยนถ้อยคำ ห้ามเรียบเรียงใหม่ ห้ามเพิ่มหรือลบเนื้อความ ห้ามเปลี่ยนการเว้นวรรค ตัวเลข หรือเครื่องหมายวรรคตอน ถ้าไม่มีคำผิดให้ตอบประโยคเดิมทุกตัวอักษร ตอบกลับเป็นประโยคที่แก้ไขแล้วบรรทัดเดียวเท่านั้น ไม่ต้องอธิบาย

ประโยค: {text}

ประโยคที่แก้ไขแล้ว:

For T2 and T3, ประโยค (‘sentence’) becomes ข้อความ (‘text’), and the single-line requirement is dropped. The English prompt of the sensitivity test below translates the T1 instruction:

Correct the misspelled Thai words in the following sentence according to standard Thai orthography. Fix only misspelled consonants, vowels, tone marks, or silent-consonant (karan) marks. Do not change the wording, do not rephrase, do not add or remove content, and do not change spacing, numerals, or punctuation. If there are no misspellings, return the sentence exactly as given. Reply with only the corrected sentence on a single line, with no explanation.

Sentence: {text}

Corrected sentence:

The few-shot prompt keeps the Thai instruction and inserts ตัวอย่าง (‘Examples’) followed by two worked examples from the train split, in the same ประโยค / ประโยคที่แก้ไขแล้ว format, before the input. The examples contain a tone-mark error (ก้อ→ก็) and a consonant-pair error (ฐาน→ฐาน):

ประโยค: แต่เพลงนี้ก็ยังไม่ได้มีการประกาศอย่างเป็นทางการว่าเป็นเพลงชาติ

ประโยคที่แก้ไขแล้ว: แต่เพลงนี้ก็ยังไม่ได้มีการประกาศอย่างเป็นทางการว่าเป็นเพลงชาติ

ประโยค: โดยถ้อยแถลงการณ์ในสมการข้างต้น เป็นฐานสอง เอนโทรปีที่จะอยู่ในหน่วยบิตเช่นกัน

ประโยคที่แก้ไขแล้ว: โดยถ้อยแถลงการณ์ในสมการข้างต้น เป็นฐานสอง เอนโทรปีที่จะอยู่ในหน่วยบิตเช่นกัน

Both variants keep the Thai system message.

Inference. Temperature is 0 where supported. Claude models use the Anthropic Message Batches API in non-thinking mode. GPT-5.x models use the Responses API, GPT-5.5 at medium or high reasoning effort with 3k and 6k output-token budgets and GPT-5.4-mini at its default effort with a 2k budget. Open reasoning models run in native thinking mode with large token budgets, and token-runout counts as a failure. Replies are cleaned by removing reasoning traces and echoed labels, keeping the last non-empty line for T1 or joining all lines for T2 and T3, and stripping quotation marks at both ends. For the 127 of 18,256 scored replies (0.7%) whose passage itself begins or ends with a quotation mark, this cleaning also removes a correct quotation mark, which counts as one error. Scoring these replies without the stripping changes any model’s correction F1 by at most 0.005 and leaves the order of models unchanged in every tier.

Prompt sensitivity. We rerun the full T1 set for one strong general model (Gemma-4-31B) and one Thai 8B model (Typhoon-S) under two alternative prompts: the same instruction in English, and the Thai instruction with two worked examples

model	Thai (default)	English	Thai + 2 examples
Gemma-4-31B	0.950	0.924	0.885
Typhoon-S 8B	0.782	0.799	0.834

Table 11: T1 correction F1 under three prompts ($n = 795$).

from the train split, which is disjoint from T1 (Table 11). Prompt choice shifts correction F1 by at most 0.07, and Gemma-4 stays ahead of Typhoon-S under every prompt. The two models move in opposite directions: the worked examples help the Thai 8B model and hurt the general model.